

Jinmin Wang (王晋民)

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Research Interests.

K -theory, index theory and noncommutative geometry

Education.

2015–2020	Ph.D. of Mathematics, Fudan University
2011–2015	Bachelor of Mathematics, Fudan University

Positions.

2024–current	Assistant Professor, Institute of Mathematics, Chinese Academy of Sciences
2021–2024	Visiting Assistant Professor, Texas A&M University
2020–2021	Research Assistant, Fudan University

Publications.

(1) (with Z. Xie) Scalar curvature rigidity of spheres with subsets removed and L^∞ metrics, to appear in *Advances in Mathematics*

(2) (with Z. Xie, G. Yu and B. Zhu) Filling Radius, Quantitative K -theory and Positive Scalar Curvature, to appear in *Geometry & Topology*

(3) (with Q. Ma, G. Yu and B. Zhu) Bounding the A-hat genus using scalar curvature lower bounds and isoperimetric constants, *Journal of Geometric Analysis*, 36, 201 (2026)

(4) (with M. Jovanovic) A scalar-mean curvature comparison theorem for manifolds with iterated conical singularities, to appear in *Communications in Analysis and Geometry*

(5) Non-negative scalar curvature on spin surgeries and Novikov conjecture, to appear in *Mathematische Zeitschrift*

(6) (with S. Cecchini, Z. Xie and B. Zhu) Scalar curvature rigidity of the four-dimensional sphere, *Math Annalen*, 394, 58. 2026

- (7) (with Z. Xie) Scalar curvature rigidity of degenerate warped product spaces. *Transactions of the American Mathematical Society*, Series B, 12(01), 1-37. 2025.
- (8) (with Z. Xie and G. Yu) A proof of Gromov's cube inequality on scalar curvature. *Journal of Differential Geometry*, 128(3), 1285-1300. 2024.
- (9) (with Z. Xie and G. Yu) Decay of scalar curvature on uniformly contractible manifolds with finite asymptotic dimension. *Communications on Pure and Applied Mathematics*, 77(1), 372-440. 2024.
- (10) (with Z. Xie, B. Zhu, and X. Zhu) Positive scalar curvature meets Ricci limit spaces. *manuscripta mathematica*, 175(3), 943-969. 2024.
- (11) (with X. Chen, Z. Xie, and G. Yu) Delocalized eta invariants, cyclic cohomology and higher rho invariants. *Annals of K-Theory*, 8(4), 531-608. 2023.
- (12) (with Z. Xie and G. Yu) Approximations of delocalized eta invariants by their finite analogues. *L'Enseignement mathématique*, 69(1), 51-91. 2023
- (13) (with Z. Xie and G. Yu) l^1 -higher index, l^1 -higher rho invariant and cyclic cohomology, *Cyclic Cohomology at 40: achievements and future prospects*, 535-579. 2021.
- (14) (with H. Liu) On localized signature and higher rho invariant of fibered manifolds. *Journal of Noncommutative Geometry*, 15(3), 919-949. 2021.

Preprints.

- (1) (with J. Wang and Z. Xie) L^∞ -metrics on tori and Schoen's conjecture, arXiv 2606.21325
- (2) (with M. Khuri, J. Wang) Riemannian positive mass theorem in all dimensions in the presence of low-codimension singularities, arXiv 2606.23526
- (3) (with P. Hochs) Quantitative partitioned index theorem and noncompact bandwidth, arXiv:2602.06666
- (4) (with Z. Xie) Scalar-mean rigidity beyond warped product spaces, arXiv:2512.05769

- (5) (with Z. Xie, G. Yu, B. Zhu) ℓ^p -coarse Baum-Connes conjecture for ℓ^q -coarse embeddable spaces, arXiv:2411.15070
- (6) (with B. Zhu and Z. Wang) Scalar-mean rigidity theorem for compact manifolds with boundary, arXiv:2409.14503
- (7) (with B. Zhu) Sharp bottom spectrum and scalar curvature rigidity, arXiv:2408.08245
- (8) (with Z. Xie) Dihedral rigidity for submanifolds of warped product manifolds, arXiv:2303.13492.
- (9) (with Z. Xie) Rigidity of strictly convex domains in Euclidean spaces, arXiv:2207.05731.
- (10) (with Z. Xie) On Gromov's flat corner domination conjecture and Stoker's conjecture, arXiv:2203.09511.
- (11) (with Z. Xie and G. Yu) On Gromov's dihedral extremality and rigidity conjectures, arXiv:2112.01510.

Invited Conference and Seminar Talks.

2026.

- Shanghai Conference on Geometry and Physics (May)

2025.

- Geometry and Convergence Program, Simons Center (September)
- Special Week on Operator Algebras 2025, Shanghai Institute for Mathematics and Interdisciplinary Sciences (August)
- 4th Australia-China-Japan-Singapore-US Index Theory Conference – Analysis and Geometry on Manifolds, National University of Singapore (August)
- Work shop on noncommutative geometry and operator algebra, Sichuan University (May)

2024.

- Texas Geometry and Topology Conference, Texas A&M University (November)
- Peking-Westlake Geometric Analysis Seminar (September, online)
- SUMIRFAS, Texas A&M University (July)
- Workshop on Non-Commutative Geometry and Higher Index Theory, University of Puerto Rico (April)
- Not Only Scalar Curvature Seminar (January, online)

2023.

- Analysis seminar, University at Buffalo (November)
- Noncommutative Geometry Festival 2023, Washington University in St. Louis (June)
- Brazos Analysis Seminar, Texas A&M University (April)
- Global Noncommutative Geometry Seminar, America (March, online)
- Joint mathematics meeting, AMS Special Session on Riemannian Manifolds with Lower Scalar Curvature Bounds, Boston (January)

2022.

- Wabash mini workshop, University of Indiana (November)
- K-theory of operator algebras and its applications to geometry and topology, University of Puerto Rico (August)
- Young Mathematicians in Noncommutative Geometry and Analysis, Texas A&M University (August)
- Recent Advances on Scalar Curvature Problems, Simons Center, Stony Brook (June)
- Great Plains Operator Theory Symposium, Washington University in St. Louis (May)
- Spring 2022 Workshop on NCG and Geometric Analysis (April, online)
- Not Only Scalar Curvature Seminar (April, online)

2021.

- Global Noncommutative Geometry Seminar, Europe (December, online)

2020.

- Workshop on Noncommutative Geometry and Quantization, Sichuan University (November)

2019.

- National Conference on Operator Theory and Operator Algebra, Zhejiang University, Hangzhou (November)
- Special Week on Operator Algebras 2019, East China Normal University, Shanghai (June)
- Noncommutative Geometry Festival 2019, Washington University in St. Louis (April)

Teaching Experience.

2025 fall	Differential Geometry
2026 spring	Functional Analysis II
2025 spring	Functional Analysis II
2023 fall	Math 251, Calculus III
2023 spring	Math 323, Linear Algebra
2022 fall	Math 251, Calculus III
2022 spring	Math 304, Linear Algebra
2021 fall	Math 251, Calculus III